

No transfers of undamaged cars should be made but the following transfers of damaged cars should be made (arriving the following day).

Glasgow to Manchester

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
3	2	3	2	3	2

Glasgow to Birmingham

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
5	5	3	4	9	5

Plymouth to Birmingham

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
5	3	6	5	5	3

The two repair depots of Manchester and Birmingham are fully occupied on all six days repairing 12 and 20 cars, respectively, each day. Repair capacity is clearly a limiting factor on the operation of the company. This is reflected by the high-shadow prices on the repair capacity constraints, which vary between £617 and £646 per car per day in both repairing depots. A plan to increase repair capacity forms the subject of problem 12.27. The solution above was obtained in 153 iterations.

14.26 Car rental 2

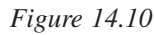
Only the Birmingham repair capacity should be increased, using both expansion options, to expand to 22 cars per day. Repair capacities in all depots are again fully used.

This allows the company to expand its fleet to 895 cars, resulting in a new weekly profit of £135 511. All demands still cannot be fully met.

This model solved in 5 nodes.

14.27 Lost baggage distribution

Two vans are needed. Solving the model, defined in Section 13.27 (a relaxation of the final model), leads to the ‘solution’ given in Figure 14.10. This required 2263 nodes to solve.



A total of 31 subtour elimination constraints needed to be appended, in 14 stages, in order to obtain a true solution with no subtours (there are a number of such solutions). This solution required 1296 nodes (a few seconds on a notepad computer). Fixing the number of vans needed at two and minimizing the maximum time needed by the vans took a further 561 nodes to solve (no additional subtour elimination constraints were needed). It produced the two routes illustrated in Figure 14.11, a bold line and a dashed line. The times taken by the vans to reach their furthest locations are 99 and 100 min, respectively.

